universidade de aveiro

deti

instituto de telecomunicações

2026/27

ApontaUA

Document Managment System



PEI GROUP 17

Our Team

Daniel Simbe
FullStack And DevOps Engineer

Anderson Tembe
FullStack Dev

Giovanni Santos
BackEnd And Architecture Engineer

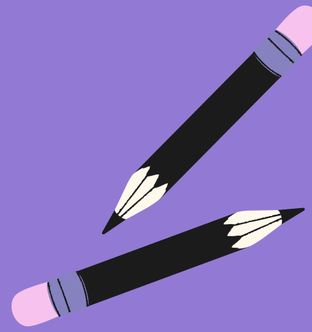
Felizardo Samuel
QA Analyst and FrontEnd Dev

Marcelo Junior
FrontEnd Dev

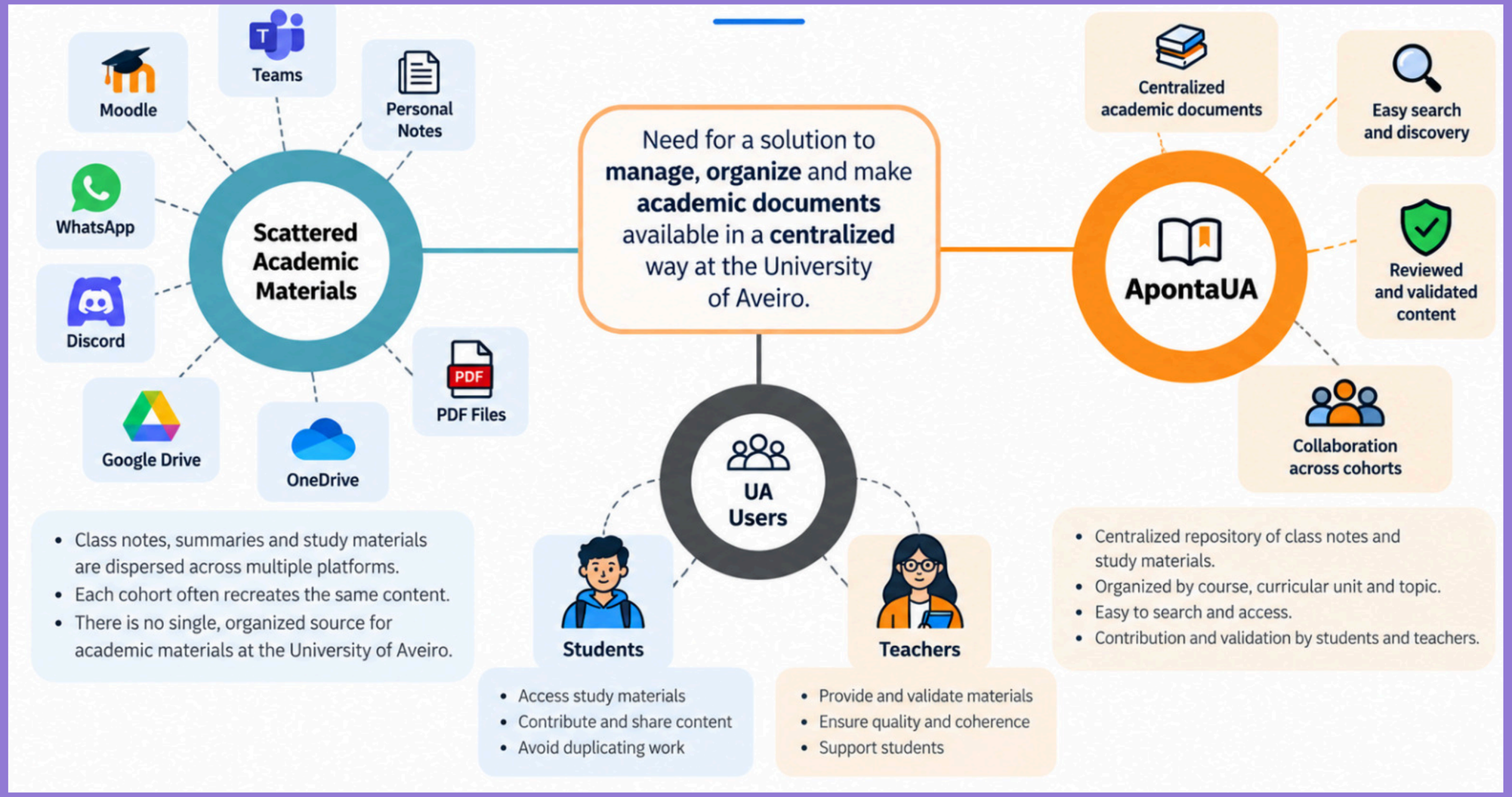
Keegan Azevedo
BackEnd Dev

João Paulo Barraca
Supervisor

Catarina Silva
Co-Supervisor



Context

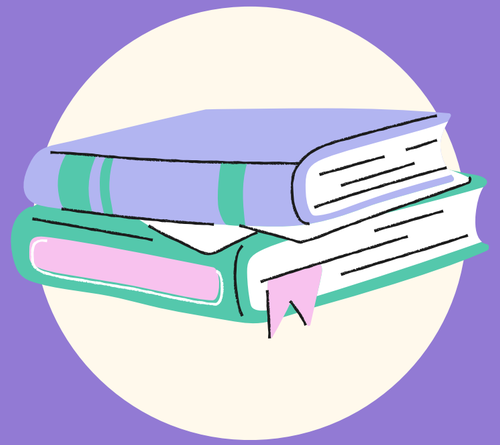

Moodle
Teams
Personal Notes
WhatsApp
Discord
Google Drive
OneDrive
PDF Files
Scattered Academic Materials
Need for a solution to manage, organize and make academic documents available in a centralized way at the University of Aveiro.
UA Users
Students
Teachers
Centralized academic documents
Easy search and discovery
ApontaUA
Reviewed and validated content
Collaboration across cohorts
• Class notes, summaries and study materials are dispersed across multiple platforms.
• Each cohort often recreates the same content.
• There is no single, organized source for academic materials at the University of Aveiro.
• Access study materials
• Contribute and share content
• Avoid duplicating work
• Provide and validate materials
• Ensure quality and coherence
• Support students
• Centralized repository of class notes and study materials.
• Organized by course, curricular unit and topic.
• Easy to search and access.
• Contribution and validation by students and teachers.

Problems



Dispersed materials

Academic notes, summaries and study materials are scattered across different places and lack a single organized source.



Duplicated effort

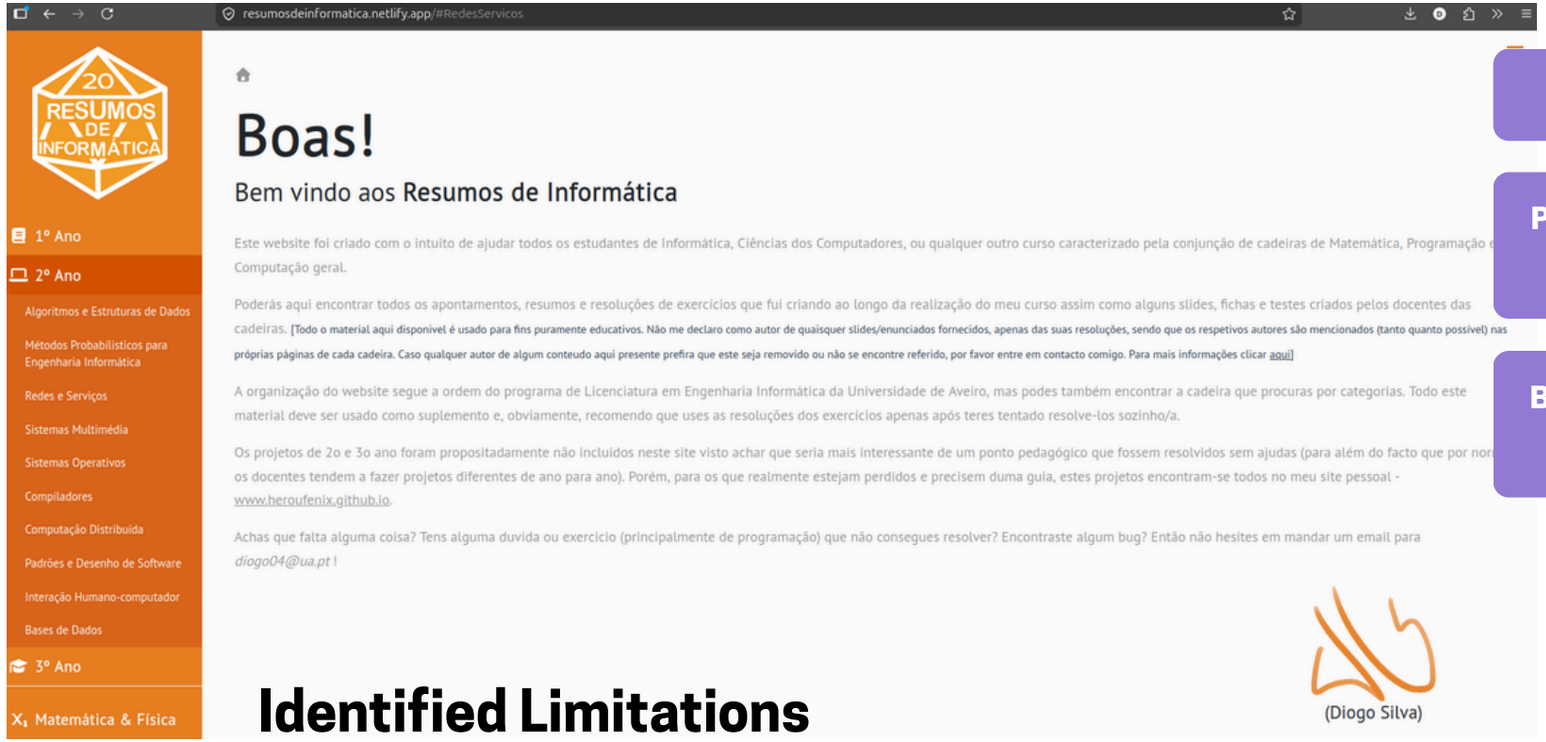
Different cohorts repeatedly recreate similar academic materials instead of reusing existing contributions.



Lack of structure and validation

There is no common structure for organizing the materials or an integrated mechanism to distinguish reviewed from non-reviewed content.

Related Work



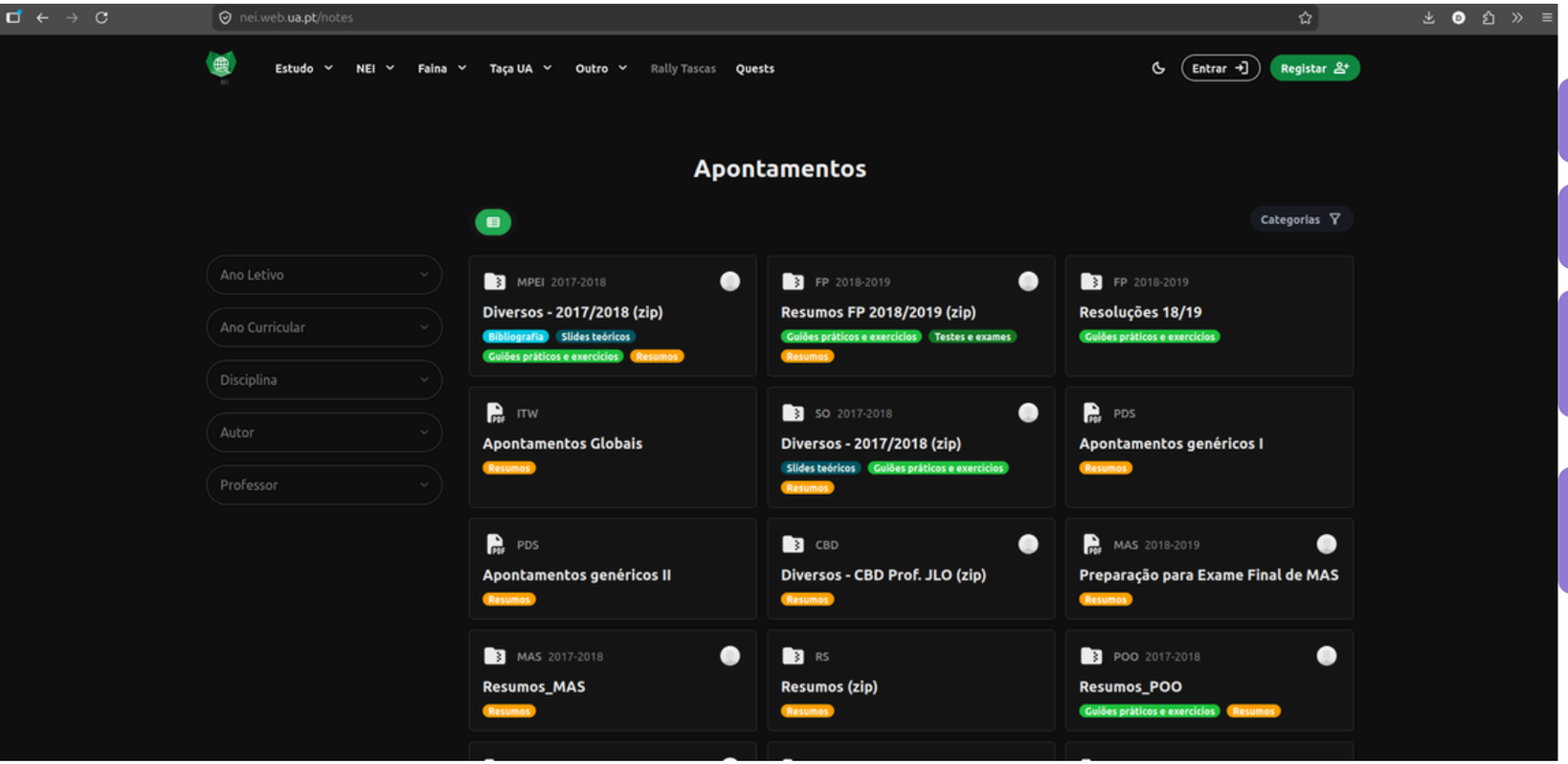
Strengths

- Organized by academic years and subjects.
- Provides notes, summaries, exercises, slides and other study materials.
- Brings materials from different courses together in one platform.

Identified Limitations

- Content is often outdated.
- Some courses are no longer part of the current curriculum.
- Some materials are duplicated.
- Limited to Computer Engineering / Informatics.

Related Work



Strengths

More modern and structured interface.

Provides more up-to-date academic materials.

Allows content to be organized and filtered through different categories.

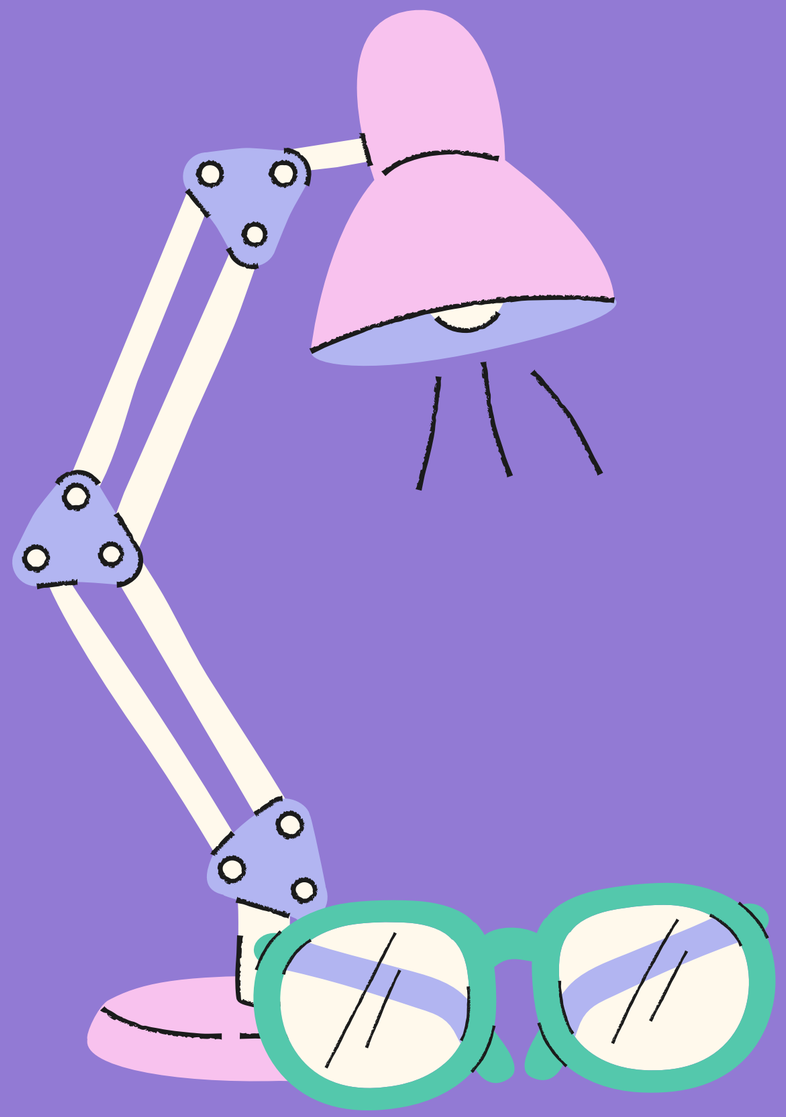
Brings academic materials and other student services together in one platform.

Identified Limitations

Materials can be scattered across different categories within the same course.

Academic content is mixed with other services, such as:

- University merchandise store.
- University championship standings.
- Other student association services.



Solution

To develop ApontaUA, a collaborative academic document management system that centralizes, organizes, and provides access to study materials, enabling their contribution, search, editing, and validation.



Goals

Centralized Academic Documents

1

- Single, organized platform for managing academic notes, summaries, and study materials.
- Documents structured by course, curricular unit, and topic.
- Easy access to materials from different cohorts.

Collaborative Content Management

2

- Students and teachers can create, contribute, edit, and share academic documents.
- Support for document versioning and collaborative content improvement.
- Encourage reuse of existing materials and reduce duplicated work.

Content Validation & Discovery

3

- Mechanisms for reviewing and validating contributed content.
- Clear distinction between reviewed and non-reviewed documents.
- Full-text search and related-content suggestions for easier discovery.

TASKS

- Requirements Analysis
- Architecture Design
- Prototype
- Development
- System Development
- Testing & Validation

EXPECTED RESULTS

- Centralized academic document management system
- Structured organization of academic materials
- Collaborative document creation and editing
- Content review and validation mechanisms
- Search and discovery of relevant materials
- Working MVP for evaluation



Communication Plan

Slack
Communication

- Communication with supervisors
- Team discussions
- Share updates and feedback
- Schedule meetings

GitHub
Code and Version Control

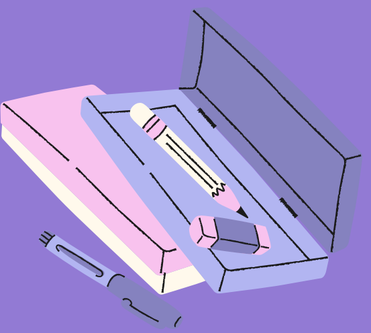
- Source code repository
- Version control using Git
- Collaborative development through pull requests
- Maintain project history and documentation

GitHub Projects
Project Management

- Organize and track tasks
- Use boards to manage the development flow
- Plan sprints and milestones
- Monitor progress

Scrum
Methodology

- Iterative and incremental development
- Organize work in sprints
- Regular team meetings (e.g., sprint planning, sprint review)
- Continuous feedback and adaptation





Project Calendar

Phase	Date	Main Tasks	Milestone
Inception	15/09/2026 – 29/09/2026	• Project website • GitHub organization • Project calendar • Context and problem definition • Goals and task planning	**M1** Lifecycle Objectives and Calendar (29/09/2026)
Elaboration	30/09/2026 – 20/10/2026	• Requirements analysis • Content model • Use cases • System architecture	**M2** Lifecycle Architecture (20/10/2026)
Elaboration (Accessibility)	21/10/2026 – 10/11/2026	• Architecture refinement • Accessibility and usability analysis • Prototype development	**MS3** Digital Accessibility and Usability (10/11/2026)
Construction	11/11/2026 – 15/12/2026	• System development (MVP) • Testing and validation • Final adjustments • Report and final presentation	**MS4** MVP and Evaluation (15/12/2026)

Thank You!

